

WENTWORTH LIU

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Data Scientist · Risk & Decision Analytics · Predictive Modelling · Applied ML

SUMMARY

Data Scientist with 2+ years of experience applying Python, SQL, statistical modelling, and machine learning to complex risk, operational, and business decisions. Built explainable risk ranking, quasi-causal analysis, NLP classification, and reusable data products for million-scale EV fleets at Li Auto; previously delivered multi-source measurement and decision analytics at ByteDance. Strong at critically validating model outputs and communicating actionable insights to technical and non-technical stakeholders.

EXPERIENCE

Li Auto Beijing, China
Data Scientist, Battery Intelligence Department Jul 2024 – Present

- **Risk Scoring & Decision Support.** Recast thermal-runaway detection under extreme class imbalance as an explainable ranking task; built a cross-platform ML pipeline with vehicle-level risk scores and feature-level drivers, achieving 93% Recall@2% on retrospectively validated high-risk cases while limiting review to the highest-risk 2% of vehicles.
- **Causal Analytics & Risk Strategy.** Used quasi-causal analysis on million-scale fleet telemetry to identify high-risk SOH-degradation cohorts and actionable drivers; productised intervention analysis across multiple vehicle models, supporting targeted strategies projected to reduce warranty exposure by tens of millions of RMB for a single model.
- **Reusable Data Products & Model Workflows.** Built a configurable feature system using custom event definitions, multi-operator aggregations, and cross-platform normalisation; deployed reusable workflows to an internal ML platform and improved feature-iteration efficiency by ~85%.
- **NLP Classification & Review Prioritisation.** Built a hybrid rule-based and statistical classifier for battery-quality work orders across 30+ fault categories, combining text matching, TF-IDF, and Random Forest; achieved Weighted F1 > 0.90 and used confidence-based routing to focus review effort on ambiguous cases.

ByteDance Beijing, China
Data Analytics Intern, Douyin Market Strategy May – Sep 2023

- **Business Measurement & Monitoring.** Integrated consumption, dissemination, and search signals across multiple data platforms into an IP popularity measurement system; added automated pipeline monitoring to address upstream data instability and support recurring content-investment and promotional decisions.
- **Segmentation & Decision Analytics.** Redesigned performance metrics and built a multi-dimensional content taxonomy for Qishui Music; diagnosed new-user conversion decline through funnel segmentation and KOL audience-saturation modelling, informing content selection and marketing-resource allocation.

Dongwu Securities China
Equity Research Intern, Mechanical Sector Dec 2021 – Mar 2022

- **Quantitative Financial Analysis.** Combined company fundamentals, competitive analysis, and market dynamics with stock-price modelling to deliver short-, medium-, and long-term investment recommendations for Harmonic Drive Systems.

EDUCATION

Nanjing University Sep 2021 – Jun 2024
MSc Computational Materials Science

- Developed Skeaf, a 3D Fermi-surface visualisation tool combining HDBSCAN clustering with quantum-oscillation data; first author, Physical Review B (2024); co-author, Physical Review B (2024) and Physical Review Materials (2023); Patent: CN115795266A.

Nanjing University Sep 2017 – Jul 2021
BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort

- National Scholarship (top 0.15%) · MCM/ICM 2020 Honorable Mention (Team Captain) · Sakura Science Exchange Programme, Tohoku University, Japan (5 selected university-wide).

SKILLS

Programming	Python · SQL/HiveSQL · C++ · MATLAB · Git
ML & Statistics	Machine Learning · Statistical Modelling · Predictive Modelling · Causal Inference · Risk Scoring · Model Evaluation · Classification · Feature Engineering · NLP · Time Series · LightGBM · scikit-learn · PyTorch
Data & Delivery	Multi-source Data Integration · Data Pipelines · Monitoring · Data Productisation · Decision Support · Visualisation · GCP · Docker · CI/CD · Spark · Hadoop · Linux
Domain & Lang.	Risk Analytics · Financial Research · Operational Analytics · English (TOEFL 104) · Mandarin/Cantonese (Native) · Japanese (JLPT N1)